

Assignment:

Module -1: Understanding of Hardware and Its Components

Section 1: Multiple Choice

1. Which of the following is NOT a component of the CPU?

1. ALU
2. RAM
3. CU
4. 1 and 3 both

→ Option 2. RAM (Right)

2. What is the function of RAM in a computer?

→ RAM (Random Access Memory) is the computer's temporary working memory. Its main function is to store data and instructions that the CPU is currently using so they can be accessed quickly.

For example:

- When you open a program, it is loaded from storage (SSD/HDD) into RAM.
- RAM helps the computer run applications smoothly and multitask efficiently.
- Data in RAM is temporary and is lost when the computer is turned off.
- In short, RAM provides fast, temporary storage for active tasks and programs.

3. Which of the following is a primary storage device?

1. HDD
2. SSD
3. SD card
4. 1 and 2 both

→ Option 4. 1 and 2 both(Right)

4. What is the purpose of a GPU?

→ A GPU (Graphics Processing Unit) is designed to process and render graphics and images quickly.

It handles tasks like:

- Displaying videos and games
- Rendering 3D graphics
- Supporting AI and machine learning computations

→ In short, the GPU improves performance for graphics-intensive and parallel-processing tasks.

Section 2: True or False

5. True or False: The motherboard is the main circuit board of a computer where other components are attached. = True
6. True or False: A UPS (Uninterruptible Power Supply) is a hardware device that provides emergency power to a load when the input power source fails. = True
7. True or False: An expansion card is a circuit board that enhances the functionality of a component. = True

Section 3: Short Answer

8. Explain the difference between HDD and SSD.
 - HDDs are cheaper and offer large storage capacity.
 - SSDs are faster, quieter, and more reliable.
9. Describe the function of BIOS in a computer system.
 - BIOS (Basic Input/Output System) is firmware stored on the motherboard that starts the computer and helps the operating system load correctly.
 - Its main functions are:
 - Performing a POST (Power-On Self-Test) to check hardware components
 - Initializing hardware such as the keyboard, hard drive, and memory
 - Loading the operating system from storage into RAM
 - Providing basic control and configuration settings for the computer.
10. List and briefly explain three input devices commonly used with computers.
 - **Keyboard**– Used to enter text, numbers, and commands into the computer.
 - **Mouse** – A pointing device used to move the cursor, click icons, and interact with items on the screen.
 - **Scanner** – Used to convert physical documents or images into digital form for storage and editing on a computer.

Section 4: Practical Application

11. Identify and label the following components on a diagram of a motherboard:
 - CPU
 - RAM slots
 - SATA connectors

- PCI-E slot

→ DONE

12. Demonstrate how to install a RAM module into a computer.

→ Turn off the computer and unplug it.

→ Open the CPU cabinet.

→ Find the RAM slots on the motherboard.

→ Open the side clips of the slot.

→ Align the RAM stick correctly and press it in firmly.

→ Make sure it clicks into place.

→ Close the cabinet and start the computer.

SECTION 5: ESSAY

13. Discuss the importance of proper cooling mechanisms in a computer.

→ Proper cooling is important in a computer because it prevents overheating. It helps the computer run smoothly, avoids damage to parts like the CPU and GPU, and improves the lifespan and performance of the system.

14. Explain the concept of bus width and its significance in computer architecture.

→ Bus width is the number of bits a computer can transfer at one time through its data bus.

Significance:

→ A larger bus width allows the computer to move more data at once, making it faster and more efficient.